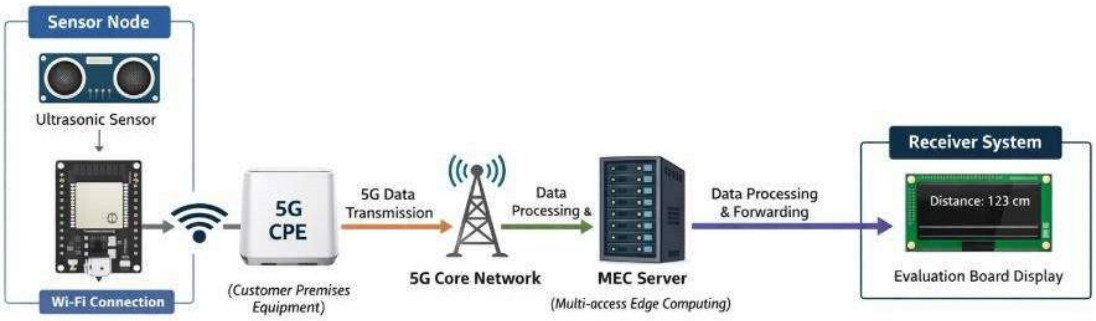
TRANSMISSION AND RECEPTION OF ESP32 SENSOR DATA USING 5G NETWORK – TEST CASE (ULTRASONIC SENSOR)

OBJECTIVES:

- To interface an ultrasonic distance sensor with the ESP32 microcontroller and acquire real-time distance measurements using embedded C programming.
- To establish Wi-Fi connectivity between the ESP32 and a 5G Customer Premises Equipment (CPE), enabling the ESP32 to transmit sensor data wirelessly.
- To utilize the 5G CPE as a gateway to the 5G core network, allowing the sensor data to be forwarded from the ESP32 into the 5G communication infrastructure.
- To transmit the sensor data from the 5G core network to a Multi-access Edge Computing (MEC) server, enabling low-latency data handling and processing.
- To forward the processed or relayed data from the MEC server to a receiver-side evaluation board, demonstrating end-to-end data transmission.

BLOCK DIAGRAM:



Transmission and Reception of ESP32 Data Using 5G Network

SYSTEM OVERVIEW:

The system enables real-time transmission of ultrasonic sensor data using an ESP32 over a 5G network. The ultrasonic sensor is interfaced with the ESP32, which connects via Wi-Fi to a 5G Customer Premises Equipment (CPE). The CPE acts as a gateway and forwards the data into the 5G core network. The sensor data is then routed to a Multi-access Edge Computing (MEC) server for low-latency handling. Finally, the data is forwarded from the MEC server to a receiver-side evaluation board, where the distance information is displayed in real time.

SOFTWARE AND COMMUNICATION FRAMEWORK:

1) Arduino IDE and Embedded C/C++ for ESP32

The Arduino IDE is used as the primary development environment for programming the ESP32 microcontroller, providing essential libraries for Wi-Fi communication and sensor interfacing. Embedded C/C++ is employed to interface the ultrasonic sensor using GPIO pins and to calculate distance based on the time-of-flight principle. The ESP32 connects to the 5G CPE Wi-Fi network and transmits the measured distance data to the MEC server (IP address: 192.168.10.12), enabling real-time sensor data communication within the 5G-enabled system.

2) UDP Communication Protocol

The User Datagram Protocol (UDP) is used for end-to-end data transmission between the ESP32, MEC server, and evaluation board. UDP enables fast, connectionless communication with minimal overhead, making it suitable for real-time sensor data forwarding where low delay is critical.

3) Python Program on MEC Server

A Python-based UDP socket application runs on the MEC server, listening on a predefined port for incoming data packets from the ESP32. The program decodes the received sensor data, performs basic validation, and forwards the extracted distance information to the receiver-side evaluation board.

4) Python Program on Evaluation Board (Receiver)

The evaluation board, running a Raspberry Pi–based operating system, executes a Python UDP receiver program. This program listens for incoming data from the MEC server and displays the received ultrasonic distance values in real time, confirming successful data reception across the 5G network.

HARDWARE FRAMEWORK:

1) ESP32 Microcontroller and Ultrasonic sensor

The ultrasonic sensor measures distance and supplies real-time data to the ESP32, which processes the information and transmits it wirelessly to the 5G CPE using built-in Wi-Fi.

2) 5G Customer Premises Equipment (CPE)

Serves as a gateway between the ESP32 and the 5G network, forwarding sensor data into the 5G core.

3) MEC Server

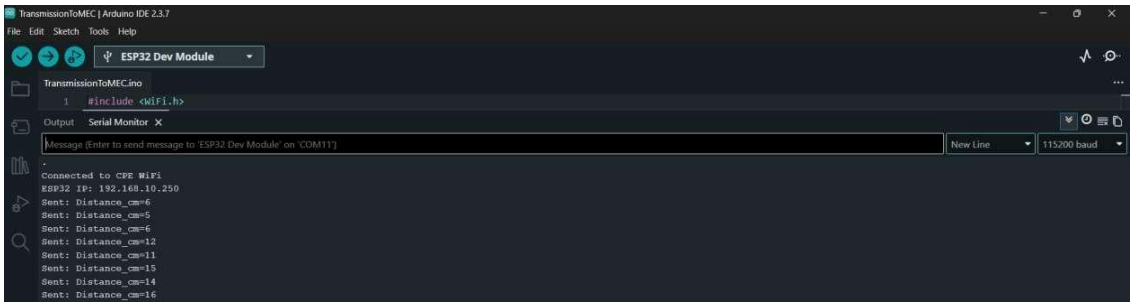
Receives data from the 5G network and forwards it to the receiver system with reduced latency

4) Evaluation Board (Receiver)

Receives and displays the transmitted distance data in real time, confirming successful end-to-end communication.

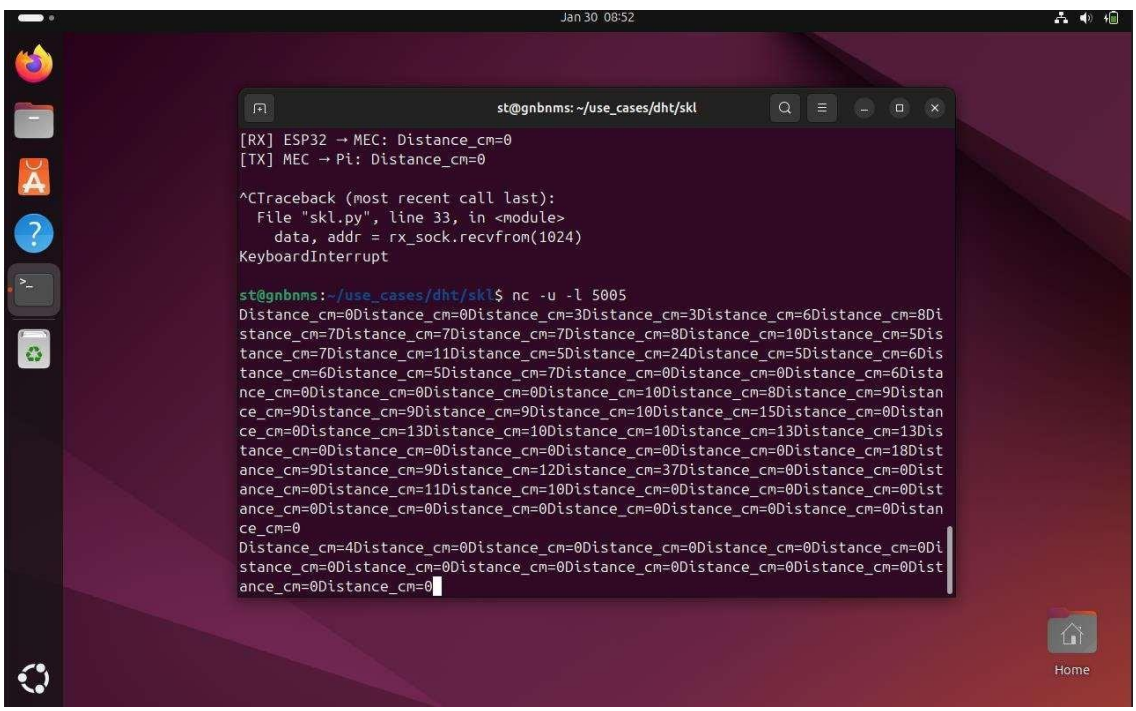
RESULTS

Step 1: ESP32 Wi-Fi based distance data transmission



The image shows the Arduino IDE serial monitor displaying real-time distance measurements sent by an ESP32 development board over Wi-Fi. The ESP32 successfully connects to a Wi-Fi network (CPE Wi-Fi) and obtains the IP address 192.168.10.250. The device continuously sends distance to the MEC server by specifying the IP address of MEC server.

Step 2: Viewing the data at MEC server



At the MEC server, the real-time sensor data transmitted from the ESP32 can be received using the command: nc -u -l 5005

nc (netcat) is a utility for reading and writing network data, -u specifies UDP mode, -l enables listening for incoming packets, and 5005 is the port on which the MEC server receives data from the ESP32. This allows MEC Server to verify the incoming data.

Step 3: Forwarding the data from MEC server to Evaluation Board at Receiver side

```
st@gnbnms: ~/use_cases/dht/skl
[TX] MEC → Pi: Distance_cm=4

[RX] ESP32 → MEC: Distance_cm=4
[TX] MEC → Pi: Distance_cm=4

[RX] ESP32 → MEC: Distance_cm=3
[TX] MEC → Pi: Distance_cm=3

[RX] ESP32 → MEC: Distance_cm=6
[TX] MEC → Pi: Distance_cm=6

[RX] ESP32 → MEC: Distance_cm=4
[TX] MEC → Pi: Distance_cm=4

[RX] ESP32 → MEC: Distance_cm=3
[TX] MEC → Pi: Distance_cm=3

[RX] ESP32 → MEC: Distance_cm=3
[TX] MEC → Pi: Distance_cm=3

[RX] ESP32 → MEC: Distance_cm=4
[TX] MEC → Pi: Distance_cm=4
```

The MEC server continuously receives real-time distance data from the ESP32 and forwards it to the receiver evaluation board. Each received measurement is validated and transmitted immediately, ensuring reliable end-to-end data flow from the sensor node through the 5G network to the evaluation system for real-time monitoring.

Step 4: Reading the data at Evaluation Board

```
aryaman@aryaman-desktop: ~/Desktop
aryaman@aryaman-desktop: ~/Desktop 80x24
aryaman@aryaman-desktop:~$ ls
Desktop    Downloads  Pictures  Templates  hydrotera    snap
Documents  Music      Public    Videos     minicom.log
aryaman@aryaman-desktop:~$ cd Desktop
aryaman@aryaman-desktop:~/Desktop$ hostname -I
10.101.1.188
aryaman@aryaman-desktop:~/Desktop$ python3 skl.py
[Pi] Waiting for data from MEC...
[RX] Distance_cm=0
[RX] Distance_cm=0
[RX] Distance_cm=0
[RX] Distance_cm=7
[RX] Distance_cm=4
[RX] Distance_cm=3
[RX] Distance_cm=4
[RX] Distance_cm=3
[RX] Distance_cm=8
[RX] Distance_cm=5
[RX] Distance_cm=0
[RX] Distance_cm=0
[RX] Distance_cm=0
[RX] Distance_cm=0
[RX] Distance_cm=7
[RX] Distance_cm=16
```

In this setup, the MEC server uses port 6000 to forward data to connected devices. The Raspberry Pi on the evaluation board listens to this port, receives the incoming data, and processes it. The processed data is then displayed on the attached interface

Conclusion

This mini project successfully demonstrates the transmission and reception of ESP32 sensor data using a 5G network. By integrating an ultrasonic sensor with ESP32, the system proves the feasibility of real-time IoT communication using next-generation wireless technology. The project highlights the potential of 5G in enabling efficient, scalable, and low-latency IoT applications.